

Research Methodology: Class1-2



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Class-1: Introduction to Research Methodology

Class-2: Formulation of the Research Problem and Development of the Research Hypotheses

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Class-1: Introduction to Research Methodology

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Class-1: Learning Objective

At the end of the class, the student must be able to understand and answer these questions:

- **What do you understand by the term research?**
- **What are the objectives of the research?**
- **What are the different types of research?**
- **What are the steps of a typical research?**
- **What are the features of a good research study?**

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How do we acquire knowledge?

Knowledge is acquired through multiple sources:

- **Tradition** – beliefs passed from generation to generation
- **Authority** – information from experts, teachers, books
- **Personal Experience** – learning through observation and practice
- **Trial and Error** – learning from mistakes
- **Logical Reasoning** – induction and deduction
- **Scientific Method** – systematic, objective, and empirical

Research Methodology is all about scientific method of learning process

What is Research? 1/2

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Noun: the systematic investigation into and study of materials and sources in order to establish facts and reach new conclusions.

"The group carries out research in geochemistry. "

Verb: investigate systematically. "She has spent the last five years researching her people's history."

The word research is derived from the Middle French "**recherche**", which means "**to go about seeking**", the term itself being derived from the **Old French term "recherchier"** a compound word from "**re-**" + "**cerchier**", or "**sercher**", meaning '**search**'. The earliest recorded use of the term was in 1577.

Seeking: to try to find or get something

What is Research? 2/2

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Research is creating new knowledge: Neil Armstrong

Research is what I am doing when I don't know what I am doing:
Wernher Von Braun

If we know what we are doing, it would not be called research:
Albert Einstein

- Research is a **systematic and scientific process** of collecting, analyzing, and interpreting data.
- It aims **to discover new knowledge** or verify existing facts.
- It **involves problem identification, hypothesis formulation, and testing, and building theories**

What are the objectives of research?

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- To gain new knowledge and insights.
- To verify and test existing theories.
- To solve specific problems.
- To establish relationships between variables.
- To support decision-making with evidence.

These are the reasons we should do research?

Types of Research: By purpose

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In research methodology, studies are categorized based on their **purpose, depth of investigation, data type, and time dimension**.

By Purpose (The Goal of the Research)

- **Exploratory Research:** Conducted when little or no prior research exists on a topic. It aims to clarify the nature of a problem (e.g., *Exploring the early psychological impacts of a newly discovered virus*). How do students feel social media affects their studies? What do teachers think about this issue?
- **Descriptive Research:** Focuses on describing characteristics of a population or phenomenon accurately and systematically (e.g., *A survey detailing the screen-time habits of teenagers*).
- **Explanatory (Causal) Research:** Seeks to identify cause-and-effect relationships between variables, answering why and how things happen (e.g., *Testing whether increasing remote work hours causes an increase in employee productivity*).

Exploratory Research – "What are the possible reasons?"

Descriptive Research – "What is happening?"

Explanatory Research – "Why is this happening?"

Types of Research: By Approach to Data

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Quantitative:
connected with
the amount or
number of
something rather
than how good it
is

Qualitative:
connected with
how good
something is,
rather than with
how much of it
there is

2. By Approach to Data (Nature of Evidence)

- **Quantitative Research:** Deals with numbers, logic, and objective data. It uses statistical and mathematical analysis to test hypotheses (e.g., *Measuring test scores to see if a new math curriculum works*).
- **Qualitative Research:** Deals with words, meanings, images, and subjective experiences. It explores opinions, behaviors, and cultures (e.g., *Conducting interviews with cancer survivors to understand their emotional journeys*).
- **Mixed Methods:** Combines both quantitative and qualitative approaches to provide a more comprehensive understanding of a research problem.

Mixed-Method Research – "Combining both approaches"

Types of Research: By Depth of Investigation

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Key Feature: Didn't
immediately solve
real-world problems
but expanded
scientific knowledge

Key Feature: The goal
was to apply scientific
knowledge to solve a
practical problem.

3. By Depth of Investigation (Outcome Goals)

- **Basic (Fundamental/Pure) Research:** Driven by curiosity and the desire to expand human knowledge without an immediate commercial or practical application (e.g., *Studying the physics of subatomic particles*).
- **Applied Research:** Designed to solve specific, practical problems facing individuals, organizations, or society (e.g., *Finding a more efficient battery design for electric cars*).

Types of Research: By Time Dimension

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Subjects at a
given time.

Over a
defined period
of time

4. By Time Dimension (How Data is Collected)

- **Cross-Sectional Research:** Observes a population at a single, specific point in time (like taking a snapshot). It is quick and inexpensive (e.g., *A survey measuring public attitudes toward artificial intelligence today*).
- **Longitudinal Research:** Follows the same subjects or variables over an extended period, sometimes years or decades (e.g., *Tracking the health outcomes of individuals who quit smoking over 20 years*).

The process of research: Step-1

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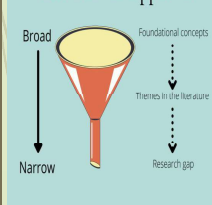
The research process is a systematic, step-by-step roadmap.

The 7-Step Research Process

Identify and Formulate the Research Problem

1. **What you do:** Pinpoint a gap in existing knowledge, a practical issue, or an unanswered question. Narrow it down from a broad topic to a specific, manageable research question.
2. **Example:** Moving from "Education is hard" to "Does gamified homework improve math retention among middle school students?"

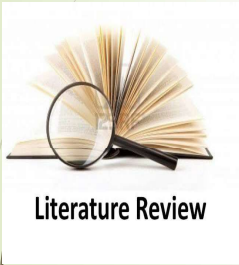
The Funnel Approach



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The process of research: Step-2

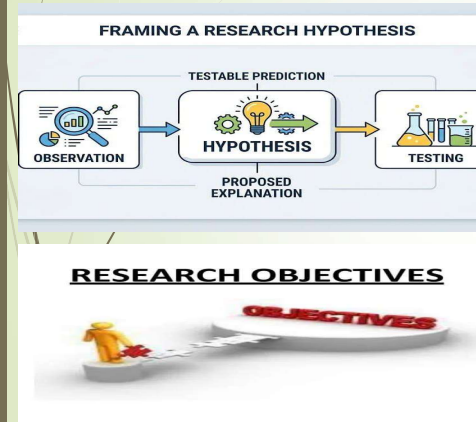
Literature Review



- **What you do:** Read existing books, academic journals, and past studies on your topic.
- **Why it matters:** It ensures you aren't reinventing the wheel, helps you find standard methodologies, and highlights what remains unknown.

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The process of research: Step-3



Formulate Hypotheses and Objectives

- **What you do:** Build tentative, testable predictions or clear study objectives that state what you expect to find or measure.

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The process of research: Step-4

RESEARCH DESIGN

Research Design & Methodology

- **What you do:** Choose your research blueprint. Will it be qualitative, quantitative, or mixed methods? Decide how you will gather data (surveys, lab experiments, interviews) and who your target population or sample group will be.

Research Methodology



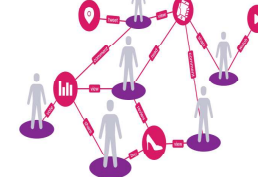
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The process of research: Step-5

DATA COLLECTION

Data Collection

- **What you do:** Execute your plan in the real world. Distribute questionnaires, run clinical trials, observe behaviors, or extract archival data. (This is often the most time-consuming phase).



The process of research: Step-6



Data Analysis and Interpretation

- **What you do:** Clean, code, and analyze your data. For quantitative data, you run statistical tests (ANOVAs, regressions); for qualitative data, you code transcripts for recurring themes. Then, you answer: *What does this data actually mean?*

The process of research: Step-7



Reporting and Publishing the Findings

- **What you do:** Write up your final research paper, thesis, or journal article. Subject it to peer review, incorporate feedback, and share your discoveries with the scientific or professional community.

Case-study: 1.1: Exercise and student

Think of a researcher who wants to know **whether daily exercise improves students' concentration:**

- Step 1: Problem** → Students seem distracted in class.
- Step 2: Literature** → Read previous studies on exercise and concentration.
- Step 3: Hypothesis** → Daily exercise improves concentration.
- Step 4: Methodology** → Select students and decide how to measure concentration.
- Step 5: Data Collection** → Record exercise habits and concentration scores.
- Step 6: Analysis** → Compare the results.
- Step 7: Reporting** → Present the conclusion.

In one line:

Problem → Literature → Hypothesis → Method → Data → Analysis → Report.

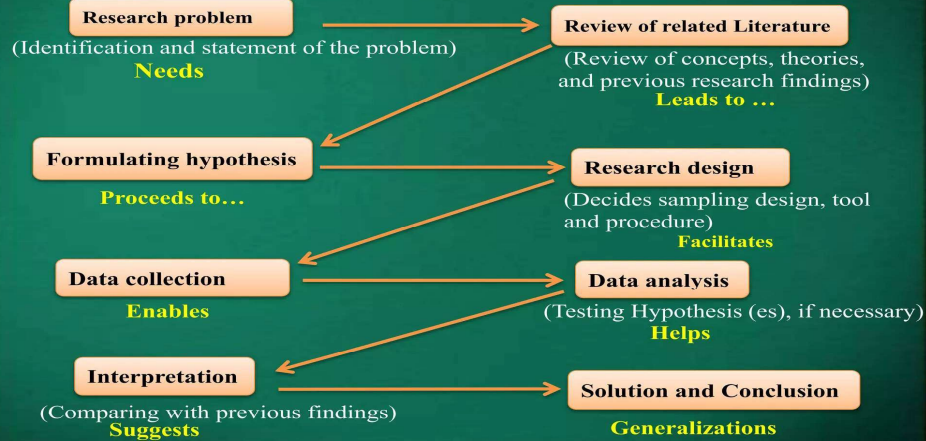
Case-study-1.2: Does Using Fertilizer Improve Plant Growth?

Imagine a farmer notices that **some plants grow faster and healthier than others**. He wonders whether fertilizer is responsible.

- Step-1. Research Problem:** He asks: "Does fertilizer help plants grow faster?"
- Step-2. Literature Review:** He reads previous studies about **fertilizers and plant growth**.
- Step-3. Hypothesis & Objectives:** He predicts: "Plants given fertilizer will grow faster than plants without fertilizer."
- Step-4. Research Design & Methodology:** He chooses **20 similar plants** and divides them into two groups:
 - Group A → Fertilizer
 - Group B → No fertilizer
- Step-5. Data Collection:** For four weeks, he records the **height of each plant every week**.
- Step-6. Data Analysis & Interpretation:** He compares the growth of the two groups and finds that the fertilizer group grew more.
- Step-7. Reporting the Findings:** He writes a report explaining the **method, results, and conclusion**.

Problem → Literature → Hypothesis → Method → Data → Analysis → Report.

Research process flow chart



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Key Features of a good research study

A good research study is defined by several core features.

- **1: Clear and Focused Research Question:** The study begins with a well-defined, testable hypothesis or research question. It avoids ambiguity and explicitly states what the research aims to discover, measure, or solve.
- **2: Rigorous Methodology:** The design matches the research question (e.g., randomized controlled trials for interventions, qualitative design for lived experiences). It uses appropriate data collection tools and analytical techniques.
- **3: Internal and External Validity:**
 - *Internal validity* ensures that the observed effects are genuinely caused by the independent variables, not confounding factors or bias.
 - *External validity* (generalizability) means the findings can be reliably applied to broader populations or real-world settings beyond the study sample.

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Key Features of a good research study

- **4: Representative and Adequate Sample Size:** The participants or subjects are carefully selected using appropriate sampling techniques (such as random sampling) to minimize selection bias, and the sample size is large enough to achieve statistical power.
- **5: Transparency and Reproducibility:** Every step of the research process, from data collection protocols to statistical code, is documented clearly. Another researcher should be able to replicate the study and verify its findings.
- **6: Ethical Integrity:** The study protects human or animal subjects through informed consent, minimizes potential harm, maintains confidentiality, and secures institutional review board (IRB) or ethical committee approval.

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Key Features of a good research study

- **7: Objectivity and Minimization of Bias:** The researchers actively control for confounding variables, researcher bias, and measurement errors through techniques like blinding, standardized data collection, and objective analytical tools.
- **8: Contribution to Existing Knowledge:** A good study doesn't just repeat past work; it addresses a clear gap in the current literature, offering new insights, making new innovations, or laying the groundwork for future inquiry.

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Case-study 1.3: "Does a Smart Classroom Really Improve Student Performance?"

- Imagine a B.Tech college introduces a new **AI-based Smart Classroom System**. The system uses AI to personalize lectures, provide instant quizzes, and recommend learning materials.
- The college principal says: **"This new AI classroom will improve students' examination performance."**
- A professor, Dr. Ananya, is curious. Instead of simply believing the claim, she decides to **conduct a research study**.

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1. Clear and Focused Research Question

Dr. Ananya does not ask: **"Is AI good for education?"**

This question is too broad.

Instead, she asks:

"Does the use of an AI-based smart classroom significantly improve the mathematics examination scores of second-year B.Tech students compared with conventional classroom teaching?"

Now the research has:

- A clear population → second-year B.Tech students
- An independent variable → AI-based classroom
- A dependent variable → examination score
- A comparison → conventional classroom
- A measurable outcome → examination performance

Feature:

A good research study starts with a clear, focused, and testable research question.

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2. Rigorous Methodology

Dr. Ananya now has to decide:

"How can I scientifically find the answer?"

She selects 200 students and divides them into two groups:

Group A → AI-based classroom

Group B → Conventional classroom

Both groups study the **same mathematics syllabus for 12 weeks**.

She ensures that:

- The same topics are taught.
- The study duration is the same.
- The final examination is the same.
- The assessment procedure is standardized.

Finally, she compares the examination scores.

Feature:

The research methodology must be appropriate for answering the research question.

The important lesson for students:

Good research does not mean collecting lots of data. It means using the RIGHT method to answer the RIGHT question.

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3. Internal and External Validity

Now Dr. Ananya gets an interesting result:

AI classroom students scored an average of **82 marks**, while conventional classroom students scored **72 marks**.

She becomes excited.

But then she asks:

"Did AI really cause the improvement?"

Maybe the AI group had better teachers.

Maybe they studied more.

Maybe they already had higher marks before the experiment.

Maybe they had better internet access.

These are **confounding factors**.

So she checks previous examination scores, teacher experience, study hours, attendance, and other important factors.

This is related to **internal validity**.

Internal Validity

"Did AI actually cause the improvement?"

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But Dr. Ananya has another question:

"Can I apply my findings to other B.Tech students?"

Her study was conducted in only one college.

Therefore, she conducts the study with students from **three different colleges**, including students with different academic backgrounds.

This improves **external validity/generalizability**.

External Validity

"Can my findings be applied beyond my study participants?"

Easy way to remember:

Internal validity = Inside the study "Is my conclusion correct?"

External validity = Outside the study "Can I apply my conclusion elsewhere?"

4: Representative and Adequate Sample Size

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Dr. Ananya initially thinks:

"I will just select 20 students who volunteer."

Her research colleague says:

"Wait! What if those 20 students are unusually motivated?"

That could create **selection bias**.

So she selects students using an appropriate sampling procedure and ensures that students from different backgrounds, branches, and academic levels are represented.

She also determines an appropriate sample size using **statistical power considerations** rather than simply choosing an arbitrary number.

Feature: A good study needs: **Right people + Right number of people**

Easy student memory:

Representative sample → "Who?"

Adequate sample size → "How many?"

5. Transparency and Reproducibility

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Dr. Ananya publishes her research.

But another researcher asks:

"How exactly did you conduct the experiment?"

Dr. Ananya provides:

- The research design
- Participant selection procedure
- AI classroom configuration
- Teaching protocol
- Questionnaire
- Examination procedure
- Data preprocessing methods
- Statistical methods
- Analysis code
- Results

Feature:

A good researcher should not say,

"Trust me."

Instead:

"Here is exactly what I did. You can check it and repeat it."

Another researcher now has enough information to **repeat the experiment**.

Suppose another college conducts the same study and obtains similar results.

That strengthens confidence in the findings.

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6. Ethical Integrity

Before beginning the experiment, Dr. Ananya asks students to participate voluntarily.

She explains:

"Your participation is voluntary. Your personal information will remain confidential, and you may withdraw from the study."

She obtains informed consent and receives approval from the appropriate institutional ethics committee.

She also ensures that:

- Students are not harmed.
- Personal information is protected.
- Academic grades are not unfairly affected.
- Students are not forced to participate.

Feature:

Good research must produce knowledge without violating the rights, safety, dignity, or privacy of participants.

7. Objectivity and Minimization of Bias

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There is one problem.

Dr. Ananya personally **believes that AI classrooms are better.**

She might unconsciously give the AI group more attention.

So she takes precautions.

For example:

- Uses the same teaching schedule.
- Uses standardized assessments.
- Uses predefined statistical methods.
- Uses blinded assessment where feasible.
- Controls important confounding variables.
- Reports both positive and negative findings.

Suppose the results finally show:

AI classroom improved performance by only **3 marks**, not the expected 15 marks.

Dr. Ananya does not change the numbers.

She reports:

"The improvement was modest and may not be educationally significant."

Feature:

A good researcher follows the evidence, even when the evidence disagrees with the researcher's expectations.

8. Contribution to Existing Knowledge

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Finally, Dr. Ananya asks: **"Why was this research necessary? Other researchers have already studied AI in education."**

She reviews previous research and discovers something interesting:

Most previous studies examined AI classrooms in universities in developed countries, but very few studies investigated their effectiveness among **Indian B.Tech students in resource-constrained institutions.**

That is the **research gap.**

Her study therefore contributes new knowledge.

She discovers:

"AI classrooms can improve performance, but the improvement depends strongly on student engagement, teacher training, and availability of digital infrastructure."

Now her study is not merely repeating old research.

It provides **new evidence and new insights.**

Feature:

Good research should answer an unanswered question, fill a knowledge gap, challenge an existing belief, or provide useful new evidence.

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Class-2: Formulation of the Research Problem and Development of the Research Hypotheses

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Learning Objective

At the end of the class the student must able to understand and answer these questions

- How **deductive thought** differs from **inductive thought** to formulate a research problem?
- How a student able to formulate his/her **research problem** from the existing literature?
- What are the basic **components of well-defined research problem**?
- What is **research hypothesis** and how hypotheses are the guiding force in any research study?
- What are the **various types of hypotheses**?

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The scientific thought: Types of reasoning

- Reasoning is the process of using existing knowledge to draw conclusions, make predictions, or construct explanations. It can be analytical (logical) or creative or mix of both.
- There are three types of reasoning, which we all use in different proportions, depending upon scenarios and our personality.
 - First one:** Deductive Reasoning or Deductive Thought
 - Second one:** Inductive Reasoning or Inductive Thought
 - Third one:** Abductive Reasoning

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Deductive Reasoning/Deductive Thought

- It is an analytical/logical reasoning approach, where we use facts or proven rules to progress from general theory to specific conclusions.
- As for example, whenever we use any mathematical equations to arrive at a certain conclusion; or reasoning like if $A > B$ and $B > C$, therefore $A > C$; or logics like if I put my finger in fire, it will burn.
- In all these examples, the conclusion are certain without any doubt. We all use deductive reasoning (consciously or sub-consciously) in our day-to-day personal and professional lives, for taking numerous decisions.

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Inductive Reasoning

- It is an analytical/logical reasoning approach, where we use historical data, patterns, associations etc. to progress from specific observation to general conclusions.
- As for example, if most of the people are complaining about the food, the food must be bad; or if an employee is consistently performing bad, he doesn't have required talent; or if price of the product reduces, sales volume increases.
- In all these examples, the conclusions are probably certain, but not 100%.
- We all use inductive reasoning (consciously or sub-consciously) in our day-to-day personal and professional lives, for taking numerous decisions.
- However, inductive reasoning goes wrong when our historical data or sample size is not sufficient, or we fail to consider some other variables.
- Most of the cognitive biases are also caused due to wrong inductive reasoning.

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Abductive Reasoning

- It is a creative reasoning approach where you progress from incomplete observation to best predictions, by making assumptions, creating hypotheses and then validating (accepting/rejecting) the same through experiments.
- Abductive reasoning is used, when the situation or problem is unique and therefore no historical data is available. The scientific discoveries made by scientists, or crime cases solved by detectives, or medical prescriptions given by doctors, uses abductive reasoning.
- In all these cases, the conclusions are merely the best possible explanation or best prediction. We all use abductive reasoning (consciously or sub-consciously) in our day-to-day personal and professional lives, but in very limited capacity.
- However, most of the complex and unique problems are solved, and innovations are created, primarily by using abductive reasoning.

Case-study:2.1: "The Case of the Broken Solar Panel"

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Three engineering students noticed that a **solar panel in their laboratory suddenly stopped producing power**. Each student used a **different reasoning method** to understand the problem.

1. Deductive Reasoning

The first student started with a **known theory**.

Theory:

If a solar panel is shaded, its power output decreases.

He checked the panel and found a **tree branch casting a shadow on it**.

Conclusion:

The shadow caused the power drop.

This is Deductive Reasoning because it moves from general theory to a specific conclusion.

Theory → Observation → Conclusion

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2. Inductive Reasoning

The second student observed several solar panels over time.

He noticed that:

- ▶ Panels under shade produced less power
- ▶ Panels receiving full sunlight produced more power

After many observations, he developed the idea:

Solar panel output strongly depends on sunlight exposure.

This is Inductive Reasoning because it moves from observations to a general conclusion.

Observation → Pattern → Theory

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3. Abductive Reasoning

The third student noticed something unusual:

The solar panel stopped working **even though there was no shade**.

He proposed a **possible explanation**:

Maybe the inverter connected to the panel is faulty.

After checking, he discovered the **inverter had indeed failed**.

This is Abductive Reasoning because it tries to find the most likely explanation for an unexpected observation.

Observation → Possible Explanation → Investigation

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Lesson for Students

Three ways to reason in research:

- ▶ **Deductive Reasoning** → Theory → Conclusion
- ▶ **Inductive Reasoning** → Observation → Theory
- ▶ **Abductive Reasoning** → Observation → Best Explanation

Simple Memory Trick

Deductive → Test a theory

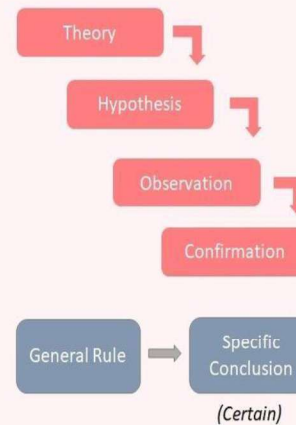
Inductive → Build a theory

Abductive → Find the best explanation

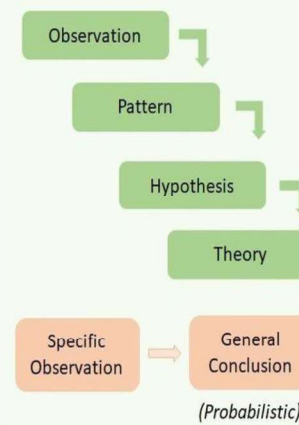
Comparison of Reasoning Methods

Feature	Deductive	Inductive	Abductive
Direction	Theory → Data	Data → Theory	Observation → Explanation
Approach	Top-down	Bottom-up	Combination
Purpose	Test existing theory	Develop new theory	Explain unexpected results
Common Research Type	Quantitative	Qualitative	Mixed research
Starting Point	Existing theory	Observations	Surprising phenomenon

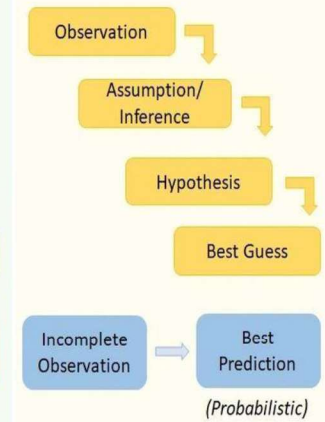
Deductive Reasoning



Inductive Reasoning



Abductive Reasoning



How to define a research problem statement?

- A research problem statement clearly defines the issue the study will examine.
- It explains the gap between the current situation and the desired situation.
- It should show who is affected, why the issue matters, and what the study will investigate.
- It must be short, focused, and easy to understand

Steps to Write a Good Problem Statement

Context

- Start with a broad topic.
- Identify the exact problem or gap.
- Explain why the problem is important.
- Mention who is affected.
- State what your study will focus on.



Problem statement = Context + Problem/Gaps + Importance + Study Focus

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Casestudy-2.3: Online classes

"Recently **online classes** are widely used. It is observed that many undergraduate students remain less engaged during virtual learning. The reasons are not well understood. This study will examine the factors influencing student engagement in online classes."

Thus, a problem statement is not just a topic title. It **identifies a real issue, shows the gap, and leads naturally to research questions.**

Can you find out a research problem in your engineering branch identifies a real issue, shows the gap, and leads naturally to research questions.

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Casestudy-2.4: Swachh Bharata Mission

Swachh Bharata Mission: Toilets are provided. In rural areas, many still avoid using it regularly. The reasons behind this low adoption remain unclear. This study will investigate the reasons: may be habit, cultural barriers, or any other that cause lesser use of home toilets."

Thus, the problem statement should be logical and easy for your audience to understand.

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Systematic Literature Review: What is a systematic literature review?

- It is a **structured method** to identify, select, and evaluate all relevant studies on a specific research question.
- It **reduces bias** by following predefined steps, inclusion and exclusion criteria, and a search strategy.
- **Planning elements** include the research question, databases, keywords, and review criteria.
- **Report the review process** clearly through a flow diagram and checklist.

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Systematic Literature Review: Steps

- Define the research topic or question.
- Set the scope of the review, such as topic boundaries, time period, and source type.
- Search for relevant literature using **databases, keywords, and references from key studies.**
- Screen and select studies using **inclusion and exclusion criteria.**
- Read, evaluate, and extract the important findings from the selected studies.
- Identify themes, patterns, gaps, and contradictions in the literature.
- **Organize the review logically**, such as thematically, chronologically, or by method.
- Synthesize the studies into a coherent summary with proper citations.

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Systematic Literature Review: Concluding remarks

Thus, literature review is not to simply state the study objectives and findings of the study under review but a **systematic evaluation of the literature under review** and **linking its relevance to the present study** as well as **highlighting research gap**. Most importantly, **proper citation of sources is mandatory** to avoid plagiarism and ethical issues in research.

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Components of a well defined problem

- There are two major types of components of a well defined problem.
- First one: The unit of analysis.
- Second one: Research Variables. The research variables are
 - Dependent variable
 - Independent variable
 - Moderating variable
 - Intervening variables
 - Extraneous variables

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The units of analysis

Definition

The unit of analysis is the person, group, organization, object, or entity about which the researcher wants to draw conclusions.

In simple words:

Unit of analysis = Who or what are you studying?

Example

Research problem:

"Does the use of AI-based learning tools improve the academic performance of B.Tech students?"

Here, the researcher is studying **individual B.Tech students**.

Therefore:

Unit of analysis = Individual B.Tech student

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Research Variables

Definition

- A **variable** is a characteristic, attribute, condition, or factor that can **take different values or change from one person, group, or situation to another**.

■ For example:

■ Age can be 18, 19, 20, etc.

■ Marks can be 45, 65, 85, etc.

■ Study hours can be 1, 3, 5 hours, etc.

- In research, variables help us examine **relationships, differences, effects, or associations**.

Example:

- *"Does the use of AI-based learning tools improve the academic performance of B.Tech students?"* contains several variables.

57 Research Variables: Independent Variables

Definition: The **independent variable (IV)** is the variable that is expected to **influence, predict, or cause a change in another variable**.

In an experiment, it is usually the variable that the researcher **manipulates or controls**.

Example: Research question:

Does the use of AI-based learning tools improve academic performance?

Here:

AI-based learning tool usage → Academic performance

Therefore:

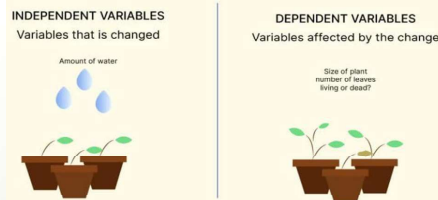
Independent Variable = AI-based learning tool usage

It could be measured as:

- No AI use
- Occasional AI use
- Frequent AI use

Easy way to remember:

Independent variable = The possible "cause" or influencing factor.



58 Research Variables: Dependent Variables

Definition

The **dependent variable (DV)** is the outcome or response that is expected to **change as a result of the independent variable**.

It is the variable that the researcher **measures**.

Example

In our study:

AI-based learning tool usage → **Academic performance**

Therefore:

Dependent Variable = Academic performance

It could be measured using:

- Examination marks
- GPA/CGPA
- Test scores
- Assignment scores

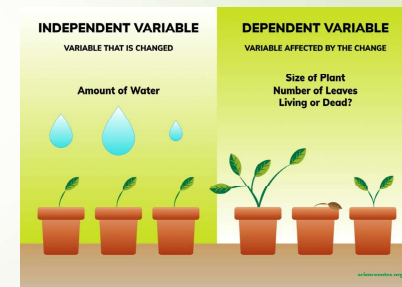
Easy way to remember:

Dependent variable = The "result" or outcome we measure.

Simple relationship:

Independent Variable → Dependent Variable

AI usage → Academic performance



59 Research Variables: Moderating Variable

Definition

A moderating variable is a variable that changes the strength or direction of the relationship between the independent and dependent variables.

In simple language:

A moderator tells us "WHEN, FOR WHOM, or UNDER WHAT CONDITIONS" the relationship is stronger or weaker.

Example

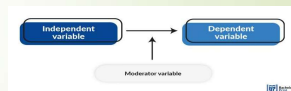
Suppose we find:

AI use → Improved academic performance

But the effect is **stronger for students who already have good digital literacy**.

Then:

Digital literacy = Moderating variable



60 Research Variables: Intervening Variable

Definition

An intervening variable explains how or why the independent variable produces an effect on the dependent variable.

It lies **between the independent and dependent variables**.

Example

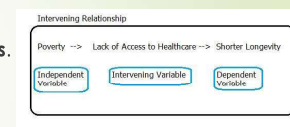
Suppose:

AI usage → Better understanding of concepts → Higher examination scores

Here, **conceptual understanding** explains how AI usage may lead to better academic performance.

Therefore:

Intervening variable = Conceptual understanding



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Research Variable: Extraneous Variable

Definition

An **extraneous variable** is a variable that is not the main focus of the research but may influence the dependent variable and interfere with the relationship being studied.

If it is not controlled, it can make the research findings misleading.

Example

We are studying:

AI usage → **Academic performance**

But students' **previous academic ability** may also affect their examination scores.

For example:

- Student A already has a CGPA of 9.0.
- Student B has a CGPA of 6.0.

Even if both use AI equally, their academic performance may differ because of their previous ability.

Therefore:

Previous academic performance = Extraneous variable

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Case-study-2.5: Student performance

Imagine two B.Tech students, **Rahul and Priya**, participate in the study.

Both use an AI learning tool.

Step 1 – Independent Variable

Rahul uses AI **frequently**.

Priya uses AI **occasionally**.

AI usage = Independent Variable

Step 2 – Dependent Variable

At the end of the semester:

Rahul scores **85 marks**.

Priya scores **72 marks**.

Academic performance = Dependent Variable

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Continue...

Step 3 – Intervening Variable

Why did Rahul perform better?

The researcher discovers that Rahul used AI to **understand difficult concepts**, which improved his conceptual understanding.

Conceptual understanding = Intervening Variable

AI usage → **Better understanding** → **Better performance**

Step 4 – Moderating Variable

But another interesting finding appears.

Students with **high digital literacy** benefit much more from AI than students with poor digital literacy.

Digital literacy = Moderating Variable

It changes **how strong the AI** → **performance** relationship is.

Step 5 – Extraneous Variable

The researcher also discovers that Rahul studied **4 hours every day**, while Priya studied only 1 hour.

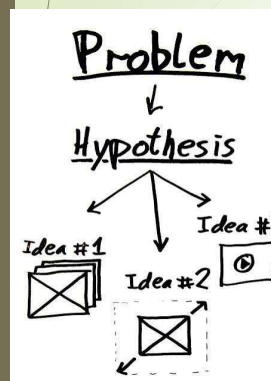
Study time could also influence examination scores.

Study time = Extraneous Variable

The researcher needs to control or account for it.

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Hypothesis: What is hypothesis



- A hypothesis is an **intellectual guess and a testable statement** about the **relationship between two or more variables**.
- It is **not same as assumption** because assumption are not be tested.
- It is a **prediction based on observation**, previous studies, or theory.
- A good hypothesis **should be clear, specific, and measurable**.

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Hypothesis: Formulation of Hypothesis

- Start with a research problem or question.
- Review related literature to find the research gap.
- Identify the independent and dependent variables.
- Write the statement in simple if-then form when possible.
- Make sure it can be tested with data.

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Hypothesis: Why is hypothesis important?

- A hypothesis **gives direction to the research and keeps the study focused.**
- It **helps choose methods, collect data, and analyze results.**
- It **connects theory with observation and makes research scientific.**

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Casestudy-2.6: Teachers evaluation

- Imagine a teacher notices that **students who revise their lessons every day usually get better marks.** (**Observation**)
- The teacher becomes curious: **"Does regular revision actually improve students' performance?"** (**Question?**)
- Based on this observation, the teacher makes a prediction: **"Students who revise their lessons regularly will score higher in examinations than students who do not revise regularly."** (**Prediction**)
- The teacher then collects data from the students and tests whether the prediction is true. (**Testing**)

This testable prediction based on an observation or existing facts is called a hypothesis.

Observation → Question → Prediction → Testing = Hypothesis in research.

Thus, a hypothesis is a clear, testable prediction. It is formed from a research question and literature review, and it guides the whole research process.

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Types of Hypothesis

There are six forms of hypothesis, and they are:

1. **Simple hypothesis**
2. **Complex hypothesis**
3. **Directional hypothesis**
4. **Non-directional hypothesis**
5. **Associative hypothesis**
6. **Casual hypothesis**
7. **Null hypothesis**
8. **Alternate hypothesis**

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Type-1: Simple Hypothesis

- It shows a relationship between one dependent variable and a single independent variable. For example – If you eat more vegetables, you will lose weight faster.
- Here, eating more vegetables is an independent variable, while losing weight is the dependent variable.
- Higher the unemployment, higher would be the rate of crime in society.
- Lower the use of fertilizers, lower would be agricultural productivity
- Higher the poverty in a society, higher would be the rate of crimes.
- Consumption of sugary drinks every day leads to obesity is an example of a simple hypothesis.

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More Example

A simple hypothesis shows the relationship between one independent variable and one dependent variable.

Example

Study hours affect student performance.

Here:

- Independent variable: Study hours
- Dependent variable: Student performance

Another example

Temperature affects battery performance.

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Type 2. Complex Hypothesis

- It shows the relationship between two or more dependent variables and two or more independent variables.
- Eating more vegetables and fruits leads to weight loss, glowing skin, and reduces the risk of many diseases such as heart disease.

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Example

Study hours, attendance, and teaching method affect student performance and examination score.

Here:

- Independent variables:** Study hours, attendance, teaching method
- Dependent variables:** Student performance, examination score

Type 3. Directional Hypothesis

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- The **directional hypothesis predicts the direction of the relationship between the independent and dependent variable.**
- A **directional hypothesis clearly indicates the direction of the relationship or difference.**
- It tells whether one variable is **higher, lower, greater, less, positive, or negative compared to another.**
- For example: children aged four years eating proper food over a five-year period are having higher IQ levels than children not having a proper meal.
- This shows the effect and direction of the effect.

Example

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- Students who study for more hours will score higher marks in exams.
- Here, **study hours** are the independent variable because they are the factor influencing the result.
- **Exam marks** are the dependent variable because they depend on how many hours the student studies.
- The direction is clearly mentioned: more study hours will lead to **higher** exam marks.

Simple Formula

Independent Variable → affects → Dependent Variable

Example:

Study hours → affect → Exam marks

Directional hypothesis:

More study hours will increase exam marks.

Type 4. Non-directional Hypothesis

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- It is a statement that a relationship exists between two variables, without predicting the exact nature (direction) of the relationship.
- **Non -directional Hypothesis predicts the relationship between the independent variable and the dependent variable but does not specific the directional of the relationship.**
- A **non-directional hypothesis** states that there is a difference or relationship, but it does not mention the direction.
- **Example- teacher student relationship influence student's learning.**
- "Caffeine causes a change in activity level" without specifying whether that change will be an increase or a decrease.

Example

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- There is a relationship between study hours and academic achievement.
- Here, **study hours** are the independent variable because they may influence the result.
- **Academic achievement** is the dependent variable because it is the outcome being measured. However, the hypothesis does not say whether more study hours will increase or decrease academic achievement. Therefore, it is non-directional.

Simple Format

There is a relationship between Independent Variable and Dependent Variable.

Example:

There is a relationship between study hours and exam marks.

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Type-5. Associative Hypothesis

- Associative hypothesis occurs when there is a change in one variable resulting in a change in the other variable.
- **Associative Hypothesis predicts an associative relationship between the independent variable and the dependent variable.**
- *When there is a change in any one of the variables, changes also occurs in the other variable.*
- **Example:** "There is a positive association between physical activity levels and overall health."

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Associative Hypothesis

An **associative hypothesis** states that two variables are related, but it does not clearly show cause and effect.

Example

There is a relationship between study hours and academic performance.

Meaning

This hypothesis only says that variables move together in some way. It does not say that study hours directly cause better marks.

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Example

Example:

There is a relationship between stress level and academic performance.

Independent Variable: Stress level

Dependent Variable: Academic performance

Here, stress and academic performance are associated, but direct cause is not strongly claimed.

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Type 6: Causal Hypothesis

- ✓ Whereas, the causal hypothesis proposes a cause and effect interaction between two or more variables.
- ✓ **Causal Hypothesis predicts a cause and effects relationship or interaction between the independent variable and dependent variable.**
- ✓ *This hypothesis predicts the effect of the independent variable on the dependent variable*
- ✓ **Example:** "Long-term alcohol use causes liver damage."

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Example

A **causal hypothesis** states that one variable causes a change in another variable.

Example

- Increasing study hours improves academic performance.

Here:

Cause: Increasing study hours

Effect: Improved academic performance

Another example

- A new teaching method improves student achievement.

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Example

A **causal hypothesis** states that one variable causes change in another variable.

Example:

Regular exercise reduces body weight.

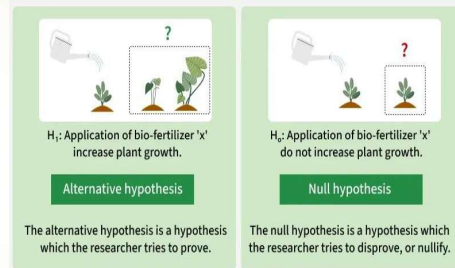
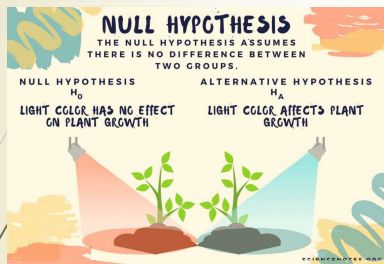
Independent Variable: Regular exercise

Dependent Variable: Body weight

Here, exercise is expected to cause a change in body weight.

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Type-7: Null Hypothesis



A simple way to remember it is: H_0 = “nothing significant is happening”; H_1 = “something significant is happening.”

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Example

Null Hypothesis

A **null hypothesis** states that there is **no significant relationship** or **no significant difference** between variables.

Example:

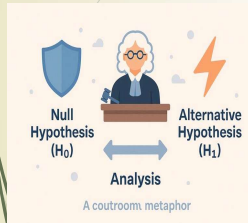
There is no significant difference in exam performance between students taught by online teaching and traditional teaching.

Independent Variable: Teaching method

Dependent Variable: Exam performance

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Type 8: Alternate Hypothesis



- An **alternative hypothesis** is a statement that says there **is a significant relationship, difference, or effect** between variables.
- It is usually written as H_1 or H_a .
- It is the opposite of the **null hypothesis**, which says there is **no significant relationship or difference**.

Definition

- An **alternative hypothesis** states that the independent variable has an effect on the dependent variable.

Simple Meaning

Something is happening.

There is an effect, relationship, or difference.

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Example

Alternative Hypothesis:

There is a significant difference in exam performance between students taught by online teaching and students taught by traditional teaching.

Independent Variable: Teaching method

Dependent Variable: Exam performance

Here, the hypothesis says that teaching method affects students' marks.

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Type-9: Statistical Hypothesis

- A **statistical hypothesis** is written in a form that can be tested using statistical methods.

Example:

- H_0 : There is no significant difference between the mean marks of Group A and Group B.
- H_1 : There is a significant difference between the mean marks of Group A and Group B.

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Example

Null Hypothesis

The null hypothesis states that there is no significant difference, no significant relationship, or no significant effect between variables.

It is denoted by:

$$H_0$$

Example

H_0 : There is no significant difference between male and female students' academic performance.

or

$$H_0: \mu_1 = \mu_2$$

Meaning

The null hypothesis assumes that any observed difference is due to chance.

Example

Alternative Hypothesis

The alternative hypothesis states that there is a significant difference, significant relationship, or significant effect between variables.

It is denoted by:

$$H_1$$

or

$$H_a$$

Example

H_1 : There is a significant difference between male and female students' academic performance.

or

$$H_1: \mu_1 \neq \mu_2$$

Meaning

The alternative hypothesis is accepted when the null hypothesis is rejected.

Type of Hypothesis	Meaning
Simple Hypothesis	Says two variables might be connected.
Complex Hypothesis	Says more than two variables might be connected.
Null Hypothesis	Says there is no connection between variables.
Alternative Hypothesis	Says there is a connection between variables.
Directional Hypothesis	Examines the specific way two variables are connected.
Non-directional Hypothesis	Says two variables are connected but doesn't say how.
Associative Hypothesis	Notes a connection but doesn't say which causes the other.
Causal Hypothesis	Says one variable directly causes another.

Thanks